ON THE SPATIALLY HOMOGENEOUS LANDAU EQUATION FOR MAXWELLIAN MOLECULES

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Abstract. We establish a simplified form for the Landau equation with Maxwellian-type molecules. We study in detail the Cauchy problem associated to this equation, and some qualitative features of the solution. Explicit solutions are also given.

Contents

1. Introduction

The Landau equation (also called sometimes Fokker-Planck) is a common kinetic model in plasma physics. It is a nonlinear partial differential equation where the unknown function, f, is the density of a "gas" in the phase space of all positions and velocities of "particles". We shall assume that these vary in $\mathbb{R}^N$, $N \geq 2$. In the case of a gas composed of a single species and if we assume that the density function is spatially homogeneous, ie does not depend on the position but only on the velocity, the Landau equation takes the following form

(1)

$$\frac{\partial f}{\partial t} = Q(f,f) = \frac{\partial}{\partial v_i}\left\{ \int_{\mathbb{R}^N} dv_*\, a_{ij}(\ v - v_*) \left[f(v_*) \frac{\partial f(v)}{\partial v_j} - f(v) \frac{\partial f(v_*)}{\partial v_{*j}} \right] \right\},$$

for $t \geq 0$, $v \in \mathbb{R}^N$, and the unknown function is assumed to be nonnegative, integrable together with its moments up to order 2. Here and below, we shall always use the convention of implicit summation over repeated indices. The matrix $(a_{ij}(z))$ is nonnegative symmetric, and depends on the interaction between particles. If we assume that any two particles at distance r interact with a force proportional to $1/r^s$, then

$$a_{ij}(z) = \Lambda |z|^{\gamma+2}(\delta_{ij} - z_i z_j/|z|^2), \tag{2}$$

with $\gamma = (s-5)/(s-1)$ for $N = 3$, and Λ is some positive constant that we shall normalize to be 1.

This equation is obtained as a limit of the Boltzmann equation, when all the collisions become grazing. See [21] for background and references on the subject. To avoid confusions with other kinetic equations, *we shall never refer to it as the Fokker-Planck equation.*

The study of the spatially homogeneous equation, besides its intrisic interest, is important for numerical applications : indeed, simulation algorithms generally make a "splitting" and consider separately the variation of f due to the gas inhomogeneities, and the one due to collisions.

A particularly simple case occurs when $s = 5$ in the three-dimensional case, or more generally $s = 2N - 1$: such molecules are the so-called "Maxwellian molecules". Although these have been intensively studied for the Boltzmann equation (see for example [3]), we are aware of very little work in that direction in the frame of Landau equation. Note that Maxwellian molecules are commonly used in numerical simulations [5, 14]. The purpose of our study is to investigate this simple case as completely as possible, in the hope to have a better understanding of the Landau equation in general. In fact, from the study of Maxwellian molecules, one can also deduce general results (see [9, 19]). A wide class of potentials is studied in other works [8, 9].

It is also of interest to compare the results obtained in the frame of the Landau equation, to those already known for the Boltzmann equation. Indeed, the Landau equation takes into account only "grazing collisions", which often entail great mathematical difficulties in the study of the Boltzmann equation – though these can generally be overcome in the homogeneous Maxwellian case [3].

In section 2 below, we shall obtain a simplified expression for the Landau equation with Maxwellian molecules, formula (9); after a brief discussion of the special case of isotropic distributions, where explicit solutions are easily available, we shall study in section 4 the form of the collision operator in the general case, then turn to the Cauchy problem associated to the equation, and insist on its regularizing properties. This will lead us to rewrite the Landau collision operator as the sum of several operators. For the convenience of the reader, this alternative form is recast in section 5, which is entirely self-contained. Sections 6 to 8 are devoted to some qualitative features of the solutions, as the decay to equilibrium, the temperature tails, and the positivity. Finally, in section 9, we exhibit a family of particular self-similar solutions.

2. Simplified Expression

We can rewrite the Landau equation as

$$\frac{\partial f}{\partial t} = \frac{\partial}{\partial v_i}\left(\overline{a}_{ij}\frac{\partial f}{\partial v_j} - \overline{b}_i f\right), \tag{3}$$

where $\overline{a}_{ij} = a_{ij} * f$, $\overline{b}_i = b_i * f$, $b_i = \partial_j a_{ij}$. It is easily checked that, at least formally, the following quantities are conserved

$$\int f(v)\,dv, \quad \int f(v)v\,dv, \quad \frac{1}{2}\int f(v)|v|^2 dv, \tag{4}$$

corresponding respectively to the mass, momentum and kinetic energy of the whole gas. We set

$$\int f = M, \quad \int fv = MV, \quad \int f|v|^2 = 2E, \tag{5}$$

where $M > 0$, $V \in \mathbb{R}^N$, $E > 0$. The equation can also be rewritten, at least formally,

$$\partial_t f = \overline{a}_{ij}\partial_{ij} f - \overline{c} f \tag{6}$$

where $\overline{c} = c * f$, $c = \partial_i b_i$. It is in this form that we shall study it. Now, in the Maxwellian case,

$$\begin{cases} a_{ij}(v) = |v|^2\left(\delta_{ij} - \dfrac{v_i v_j}{|v|^2}\right) = |v|^2\delta_{ij} - v_i v_j, \\[2ex] b_i(v) = -(N-1)v_i, \\[2ex] c(v) = -N(N-1). \end{cases}$$

Writing $f_* = f(v_*)$, we find

$$\overline{a}_{ij} = \int |v-v_*|^2 \delta_{ij} f_* \, dv_* - \int (v-v_*)_i (v-v_*)_j f_* \, dv_*$$

$$= \delta_{ij} \left(M|v|^2 + 2E - 2M(V \cdot v) \right) + M(V_i v_j + V_j v_i - v_i v_j) - \int f v_i v_j,$$

$$\overline{b}_i = -(N-1) \int (v-v_*)_i f_* \, dv_* = -(N-1) M (v_i - V_i),$$

$$\overline{c} = -N(N-1)M.$$

Thus, we are led to compute all second moments of f. Setting

$$B_{ij} = \int f v_i v_j, \tag{7}$$

we have (at least formally)

$$\frac{d}{dt} B_{\alpha\beta} = -\int \left(\overline{a}_{ij} \partial_j f - \overline{b_i} f \right) \partial_i (v_\alpha v_\beta)$$

which, after elementary computation, integrating systematically by parts, yields the ordinary differential equation

$$\frac{d}{dt} B_{\alpha\beta} = 8\delta_{\alpha\beta} ME + 4M^2 \left(N V_\alpha V_\beta - \delta_{\alpha\beta} |V|^2 \right) - 4NMB_{\alpha\beta},$$

the solution of which is

$$B_{\alpha\beta}(t) = B_{\alpha\beta}(\infty) - \Big(B_{\alpha\beta}(\infty) - B_{\alpha\beta}(0) \Big) e^{-4NMt}.$$

For example, if $V = 0$, $B_{\alpha\beta}(\infty) = (2E/N)\delta_{\alpha\beta}$. We note that the evolution of the moments of order 2 is entirely determined by the "gross conditions" M, V, B_{ij}, of f_0. This is also true for the Boltzmann equation [11]. Thus, in turn, the coefficients $\overline{a}_{ij}, \overline{b}_i, \overline{c}$, depend only on these quantities.

Now, for the sake of simplicity, we assume without loss of generality that $V = 0$, and we change the unknown function f to $\tilde{f}$ such that $f = M\alpha^{-N} \tilde{f}(./\alpha)$, $\alpha^2 = 2E/(MN)$. Then, calling the new function f, we have

$$\int f = 1, \qquad \int f v = 0, \qquad \int f |v|^2 = N, \tag{8}$$

and the corresponding equilibrium distribution is simply the Maxwellian with temperature 1,

$$\mathcal{M}(v) = \frac{e^{-|v|^2/2}}{(2\pi)^{N/2}},$$

while $B_{ij}(\infty) = \delta_{ij}$. Finally we rescale time : $\tilde{t} = (N-1)t$. Thus, equation (1) simply reduces to

$$(9) \qquad \begin{cases} \partial_t f = \overline{a}_{ij}\partial_{ij} f + Nf \\ \overline{a}_{ij} = \delta_{ij} + \dfrac{1}{N-1}\left(|v|^2\delta_{ij} - v_i v_j\right) - e^{-\alpha t} D_{ij} \\ \alpha = \dfrac{4N}{N-1}, \quad D_{ij} = \dfrac{1}{N-1}\left(\displaystyle\int f_0 v_i v_j - \delta_{ij}\right). \end{cases}$$

This equation is *linear*. It is equivalent to the nonlinear Landau equation only if we consider those solutions that have mass 1, bulk velocity zero, and energy $N/2$. We mention that this expression was obtained independently (and simultaneously) by Lemou [14] (for $N = 3$).

Before going further, we shall study the simple isotropic case.

3. The isotropic case : Fokker-Planck equation

Assuming that f is radially symmetric, and satisfies the normalization conditions (8), we have

$$\int f v_i v_j = \delta_{ij}, \qquad D_{ij} = 0.$$

We set $f = \varphi(x)$, $x = |v|^2/2$. Thus

$$\frac{\partial f}{\partial v_i} = v_i\varphi'(x), \quad \frac{\partial^2 f}{\partial v_i \partial v_j} = v_i v_j \varphi''(x) + \delta_{ij}\varphi'(x),$$
$$\Delta f = 2x\varphi''(x) + N\varphi'(x),$$
$$\sum_{i,j} v_i v_j \frac{\partial^2 f}{\partial v_i \partial v_j} = 4x^2\varphi''(x) + 2x\varphi'(x),$$
$$\sum_i v_i \frac{\partial f}{\partial v_i} = 2x\varphi'(x)$$

and we notice that

$$|v|^2\Delta f - \sum_{i,j} v_i v_j \frac{\partial^2 f}{\partial v_i \partial v_j} = 2(N-1)x\varphi'(x) = (N-1)\left(\sum_j v_j \frac{\partial f}{\partial v_j}\right).$$

In that case the Landau equation can be rewritten

$$\frac{\partial f}{\partial t} = Lf = \Delta f + v \cdot \nabla_v f + Nf = \nabla \cdot (\nabla f + fv)\,.$$

This is the so-called linear Fokker-Planck (or Ornstein-Uhlenbeck) equation, well-known in kinetic theory as well as in probability, and associated with the so-called Ornstein-Uhlenbeck adjoint semigroup (its adjoint is $\Delta_v - v \cdot \nabla_v$, associated to the standard Dirichlet form for the Gaussian measure). It can be used to describe the relaxation of Brownian molecules in a gas. Its appearance in topics related to the Boltzmann equation was already noticed by Bobylev [2, 3] : in particular, this equation is satisfied by Bobylev's explicit solutions. (We point out that in these papers the Fokker-Planck operator does not appear as a special case of the Landau equation, but is related to the invariance properties of the Boltzmann equation.)

The occurrence of the Fokker-Planck equation is not surprising, in that it is one of the most simple linear equations for which conservation of nonnegativity, mass, energy and decrease of entropy hold (this last one due for example to the Heisenberg inequality applied to $\sqrt{f}$).

The explicit solution of this equation is well-known and can be obtained readily by Fourier-transform techniques :

$$\left\{ \begin{aligned} f(v) &= \frac{e^{-\frac{|v|^2}{2\delta}}}{(2\pi\delta)^{N/2}} * e^{Nt} f_0(e^t \cdot), \\ \\ \delta &= 1 - e^{-2t}. \end{aligned} \right.$$

From this expression the asymptotic behaviour is clear : for $t \to \infty$ the first term of the convolution product converges to $\mathcal{M}$, while the second term converges to a Dirac mass. We note that the Ornstein-Uhlenbeck adjoint semigroup is commonly used to obtain smooth interpolations between an arbitrary density and its associated Maxwellian (see [6] for an application in the field of kinetic theory).

In [14] Lemou obtained a different expression of radial solutions of the spatially homogeneous Landau equation, in the form of a convergent series : it is a tedious but simple matter to check that the two expressions are equivalent provided that the convergence of all expressions is ensured. In particular, taking as initial data, for $N = 3$,

$$f_0(v) = \mathcal{M}(v) \left(1 + \frac{|v|^4}{120} - \frac{|v|^2}{12} + \frac{1}{8} \right),$$

we recover the simplest of Lemou's solutions,

$$h(t,v) = \mathcal{M}(v)\left(1 + e^{-4t}\left(\frac{|v|^4}{120} - \frac{|v|^2}{12} + \frac{1}{8}\right)\right).$$

The expressions obtained by Lemou are well adapted to treat initial data of the type (Maxwellian $\times$ polynomial), while the one given above is more convenient for arbitrary initial data, which is compensated by a much greater cost in computation (the expression is only semi-explicit). We indicate briefly a systematic method to change from one expression to the other. It suffices for that purpose to compute convolution products of the form $\mathcal{M}_1 * P(v)\mathcal{M}_2$, P being a polynomial in v and $\mathcal{M}_1, \mathcal{M}_2$ Maxwellian distributions. These computations are most easily done in Fourier representation, where they take the form $\mathcal{M}^1 P(D)\mathcal{M}^2$, where D stands for the derivation operator, $\mathcal{M}^1$ and $\mathcal{M}^2$ being Maxwellian distributions. Expanding this expression, we obtain sums of terms of the form (Maxwellian $\times$ polynomial). Then, we come back in the usual representation, having only to compute derivatives of Maxwellians.

Remark 1. We note that Bobylev's family of invariant solutions [2] consists precisely in those obtained by applying the adjoint Ornstein-Uhlenbeck semigroup to initial data of the form $\mathcal{M}|v|^2$ (up to a normalization). They satisfy the Boltzmann equation with Maxwellian potential,

$$\partial_t f = Q(f,f) = Q^+(f,f) - \beta f, \tag{10}$$

$$Q^+(f,f) = \int_{S^{N-1}\times\mathbb{R}^N} d\omega\, dv_*\, b(\alpha) f(v') f(v'_*),$$

with $v' = (v - (v - v_*, \omega\omega)$, $v'_* = (v_* + (v - v_*, \omega)\omega)$, $\cos\alpha = (\frac{v-v_*}{|v-v_*|}, \omega)$, and b is a nonnegative function (or measure) such that $\int d\omega\, b(\alpha) = \beta$. Using a spherical change of variables with axis $v - v_*$ as in [18], we write $b(\alpha)\, d\omega = \zeta(\theta)\, d\theta\, d\phi$, with $\alpha = \pi - 2\theta$. Then

$$\mu = \int \zeta(\theta) \sin^2\theta\, d\theta$$

is, up to a universal constant, the first eigenvalue of the linearized collision operator, and the only parameter upon which these solutions depend. Therefore, they are invariant under the asymptotics of grazing collisions [21] if one chooses a sequence of cross-sections

$$\zeta_\varepsilon(\theta) = \frac{1}{\varepsilon}\,\zeta\left(\frac{\theta}{\varepsilon}\right)\frac{\sin^2\left(\frac{\theta}{\varepsilon}\right)}{\sin^2\theta} 1_{\theta\le\varepsilon\pi/2}.$$

It can be easily checked that (apart from the equilibrium distribution) they are the only solutions of the Fokker-Planck equation taking the form $M(v)P(v)$, where M is a Maxwellian distribution and P a collisional invariant. We think it likely that they constitue the only simple class of solutions invariant by the asymptotics of grazing collisions.

Finally, we note that Bobylev has exhibited a countable family of radial semi-explicit solutions (in the form of a convergent series) that check both Fokker-Planck and Boltzmann equation for some particular values of the parameter μ.

Remark 2. In one dimension of velocity-space, the Landau-type model corresponding to the Kac equation is simply

$$\partial_t f = \partial_{xx} f + x\partial_x f + f = \partial_x \left(\partial_x f + xf\right),$$

together with the conditions $\int f = 1 = \int fx^2$.

After this brief discussion we go back to the general anisotropic case.

4. The collision operator and the Cauchy problem

The matrix $\overline{a}_{ij}$ is clearly nonnegative in view of its definition. We analyse its positivity. We can write

$$\overline{a}_{ij} = \left\{\delta_{ij} - e^{-\alpha t} D_{ij}\right\} + \frac{1}{N-1}\left\{|v|^2\delta_{ij} - v_i v_j\right\} \equiv \mathcal{A}_{ij} + \mathcal{B}_{ij} \tag{11}$$

The matrix $(\mathcal{B}_{ij})$ is obviously nonnegative in view of the Cauchy-Schwarz inequality; we note that it is simply (up to a constant factor) $|v|^2$ times the orthogonal projection on the space orthogonal to v. At each non-zero velocity v it has only one degenerate direction, namely that of v (so that when applied to radial functions, this operator is totally degenerate and reduces to a first-order operator, as we have seen in the preceding section).

As for the first part :

$$\begin{aligned}(N-1)\mathcal{A}_{ij} &= N\delta_{ij} - e^{-\alpha t}B_{ij} - (1-e^{-\alpha t})\delta_{ij}\\ &= (N-1)(1-e^{-\alpha t})\delta_{ij} + e^{-\alpha t}(N\delta_{ij} - B_{ij}).\end{aligned}$$

To prove that the matrix $(\mathcal{A}_{ij})$ is nonnegative, we just have to prove that

$$(B_{ij}) \leq N(\delta_{ij})$$

in the sense of matrices. This is an easy matter in view of

$$\sum_{ij} B_{ij}\xi_i\xi_j = \int \sum_{ij} f(v)v_iv_j\xi_i\xi_j = \int f(v\cdot\xi)^2 \leq |\xi|^2 \int f|v|^2 = N|\xi|^2.$$

The degeneracy can only happen if there is always equality in Cauchy-Schwarz inequality, that is, if f is concentrated on some ξ-axis.

Thus we have the following

Proposition 1. *The second-order operator $\sum \overline{a}_{ij}\partial_{ij}$ is the sum of two elliptic operators ;*
(i) the first one has coefficients depending (smoothly) only on t and is always strictly elliptic unless $t = 0$ and f_0 is concentrated on a single line, whose direction is the only degeneracy direction;
(ii) the second one has coefficients depending only on v, reduces to 0 for $v = 0$ and elsewhere has always one unique degenerate direction, namely that of v.

As we shall always assume $f \in L^1$, we shall never be concerned with the degeneracy indicated in part 1 of this proposition. However, it would be possible to assume only that f is a nonnegative measure with bounded mass and energy.

We can now rewrite the equation as

$$\partial_t f = \mathcal{L}f = \mathcal{L}_1 f + \mathcal{L}_2 f, \tag{12}$$

where

$$\begin{cases} \mathcal{L}_1 f = \displaystyle\sum_{ij} \mathcal{A}_{ij}\partial_{ij} f + \sum_i v_i \partial_i f + Nf, \\ \\ \mathcal{L}_2 f = \displaystyle\sum_{ij} \mathcal{B}_{ij}\partial_{ij} f - \sum_i v_i \partial_i f. \end{cases}$$

This decomposition provides an easy interpretation of the equation. Indeed, suppose for simplicity that $(D_{ij}) = 0$: then $\mathcal{L}_1$ is the usual Fokker-Planck operator (that we shall denote by L). As for $\mathcal{L}_2$, we can make its structure more clear by a change to polar coordinates. To obtain practical explicit expressions, we shall first illustrate this in the cases $N = 2$ and $N = 3$. Writing $f(v_1, v_2) = f(r, \theta)$, where $r \geq 0$, $v_1 = r\cos\theta$, $v_2 = r\sin\theta$, we obtain, after some computation, the following simple expression.

Proposition 2. *For $N = 2$, in polar coordinates,*

$$\mathcal{L}_2 f = \partial_{\theta\theta} f.$$

Moreover, if $(D_{ij}) = 0$, then

$$\mathcal{L}_1 f = \partial_{rr} f + \frac{1}{r^2}\partial_{\theta\theta} f + \left(\frac{1}{r} + r\right)\partial_r f + 2f$$

Thus we see that the semigroup associated to $\mathcal{L}$ consists in a Fokker-Planck diffusion process, plus an additional diffusion on each centered sphere. This second diffusion is very important for large r, where it strongly prevails over the spherical diffusion induced by the Laplace operator.

We also remark from their expression in polar coordinates that these two operators commute. But we know explicit formulas for the associated semigroups; indeed we have

$$e^{t\mathcal{L}_2} f_0 = \sum_{n=-\infty}^{+\infty} c_n(f_0, r) e^{-n^2 t} e^{in\theta} \tag{13}$$

where

$$c_n(f,r) = \frac{1}{2\pi} \int f(r,\theta) e^{-in\theta} d\theta.$$

Thus, denoting by M_δ the centered Maxwellian with mass 1 and temperature δ,

$$e^{t\mathcal{L}} f_0 = e^{t\mathcal{L}_1} e^{t\mathcal{L}_2} f_0 = M_\delta * e^{2t} \left[\left(e^{t\mathcal{L}_2} f_0 \right) (e^t \cdot) \right] =$$

$$\frac{1}{(2\pi\delta)} \int r_* dr_* d\theta_* e^{-r^2/2\delta} e^{-r_*^2/\delta} e^{r r_* \cos(\theta-\theta_*)/2\delta} e^{2t} \sum_{n=-\infty}^{+\infty} c_n(f_0, e^t r_*) e^{-n^2 t} e^{in\theta_*}$$

The convergence of the series in (13) is absolute for almost all r and all $t > 0$, and must be taken in distributional sense for $t = 0$. To justify the expression above it suffices to note that $\mathcal{L}_2$ preserves the positivity and the integrability of its solution.

From the qualitative point of view, we notice that the solution becomes immediately very regular (C^∞) and that it has a tendency to *become radial very quickly*, due to the important spherical diffusion.

For $N = 3$, we use the usual spherical coordinates, that is, $v_1 = r \sin\theta \cos\phi$, $v_2 = r \sin\theta \sin\phi$, $v_3 = r\cos\theta$. We recall that

$$\Delta = \Delta_r + \frac{1}{r^2} \Delta_{\theta,\phi},$$

where

$$\Delta_r f = \frac{1}{r^2} \partial_r (r^2 \partial_r f),$$

$$\Delta_{\theta,\phi} f = \frac{1}{\sin\theta} \partial_\theta (\sin\theta\, \partial_\theta f) + \frac{1}{\sin^2\theta} \partial_{\phi\phi} f,$$

that is, $\Delta_{\theta,\phi}$ is the Laplace-Beltrami operator.

It is possible to check

Proposition 3. *If $N = 3$, in spherical coordinates,*

$$\mathcal{L}_2 f = \frac{1}{2} \Delta_{\theta,\phi} f.$$

Moreover, if $(D_{ij}) = 0$*, then*

$$\mathcal{L}_1 f = \Delta_r f + \frac{1}{r^2}\Delta_{\theta,\phi} f + r\partial_r f + 3f.$$

Again we have exactly the same interpretation : a very strong spherical diffusion that makes f become radial quickly. It is possible to obtain a formula analogous to the above ones : we just have to expand f in spherical harmonics (which are eigenfunctions of the Beltrami operator) to get an explicit solution of $e^{t\mathcal{L}_2} f$.

More generally, in any dimension, $\mathcal{L}_2$ can be interpreted as a diffusion on the $(N-1)$-dimensional sphere. Indeed, recalling the definition of the Laplace-Beltrami operator, we fix some velocity $v_0 \neq 0$, and set

$$g(v) = f\left(|v_0|\frac{v}{|v|}\right),$$

what we need to prove is that

$$(N-1)\mathcal{L}_2 f(v_0) = |v_0|^2 \Delta g(v_0).$$

This is an easy exercise in differential calculus. It can then also be checked that $\mathcal{L}_1$ and $\mathcal{L}_2$ commute for $(D_{ij}) = 0$.

By the way, we mention that Lemou [14] has noticed that the assumption $(D_{ij}) = 0$ makes it possible to obtain semi-explicit solutions in particular cases.

We can now examine the case $(D_{ij}) \neq 0$. First we notice that we can simplify the matrix (D_{ij}) by a rotation : indeed,

$$q : \vec{e} \in \mathbb{R}^N \longmapsto \int f(v,\vec{e})(v,\vec{e})\, dv$$

defines a nonnegative symmetric bilinear quadratic form, so that there exists an orthonormal basis $(\vec{e_1}, \ldots, \vec{e_N})$ which is also orthogonal for q. In this basis, the isotropic Fokker-Planck and Laplace-Beltrami operators are clearly unchanged, while (D_{ij}) is simply $\mathrm{Diag}(D_{11}, \ldots, D_{NN})$, with $(N-1)D_{ii} = \int f_0 v_i^2 - 1$, so that, of course, $\sum_i D_{ii} = 0$. The interpretation is easy : if the "energy along the i-axis", $T_i = \int f v_i^2/2$, is too small compared with the equilibrium value, then the diffusion is reinforced in that direction, so that this energy increases; on the contrary, the diffusion is slowed down in the direction i if $\int f v_i^2 > 1$.

Thanks to a linear change of variables (with coefficients depending on t) it is still possible to write $\mathcal{L}_1$ as a Fokker-Planck operator. We just have to consider (α_{ij}) the nonnegative symmetric square root of the matrix $(\delta_{ij} - e^{-\alpha t} D_{ij})$ and take as new variables the column vectors of the inverse of this matrix : that is, in our well-chosen basis, simply

$$v_i = \sqrt{1 - e^{-\alpha t} D_{ii}}\, \tilde{v}_i$$

(or, in short, $v = S \cdot \tilde{v}$).

Suppose g is a solution of $\partial_t g = \mathcal{L}_1 g$. In the new variables we have, setting $\tilde{g}(\tilde{v}) = g(v)$,

$$\partial_t \tilde{g} = S'(t) v \cdot \partial_{\tilde{v}} \tilde{g} + L\tilde{g},$$

where L stands for the classical Fokker-Planck operator and $S_i' = \partial_t (1 - e^{-\alpha t} D_{ii})^{-1/2} = -1/2(\alpha e^{-\alpha t} D_{ii})^{-3/2} (1 - e^{-\alpha t} D_{ii})^{-3/2}$ (note that $v \cdot \nabla f$ remains unchanged).

Thus $\mathcal{L}_1$ can be considered as a "distorted" Fokker-Planck operator. Unfortunately, $\mathcal{L}_1$ and $\mathcal{L}_2$ will not commute any more ; using $\sum C_{jj} = 0$, we find

$$[\mathcal{L}_2, \mathcal{L}_1] = -\frac{e^{-\alpha t}}{N-1} \sum_{ij} D_{jj} \Big(4(v_j \partial_i - v_i \partial_j) \partial_{ij} - 2\partial_{jj} \Big).$$

So no simple explicit formula is *a priori* available.

After these preparations, we can investigate the Cauchy problem associated to our equation. It was proven by Smirnova [16] under quite general hypotheses that a parabolic equation whose coefficients behave at infinity like $|v|^2$ (this is the limit case considered) admits a unique classical solution (also unique in a class of suitable generalized solutions). In the present case, we can directly recover this theorem in a number of ways. For instance, we remark that a weak solution can be obtained as a limit of solutions of the Boltzmann equation [21]. By the way, uniqueness can also be proven by the nonexpansivity of the metric introduced in [18]. We can also construct approximate solutions by truncating the coefficients at infinity, and then pass to the limit. Local regularity is then clear since the equation is locally strictly elliptic with C^∞ coefficients : thus we get a classical solution. This is true even if f_0 is not regular : in all cases, $\mathcal{L}$ becomes strictly elliptic for all $t > 0$.

In fact the study made above enables us to conclude that there is *global regularization* of the solution, under the only hypothesis that $f_0 \in L^1$. Indeed, this is clear in the case when we have an explicit solution, and can be proven in the general case by considering the equation in polar variables : there we have a strictly elliptic equation with smooth bounded coefficients for $r \to \infty$, on $\mathbb{R} \times S^{N-1}$, where S^{N-1} stands for the $(N-1)$-dimensional sphere (the singularity at the origin is nonessential).

Of course, this solution will be a solution of the Landau equation (1) corresponding to (9) only if f_0 checks condition (8); more generally, we have to restrict to those f_0 which have bounded mass and energy, ie, more or less, those that make sense from the physical point of view.

We sum up with the

Proposition 4. *For each nonnegative f_0 having finite mass and energy, the Landau equation (1) admits a unique classical solution $f(t,v)$, defined for all $t \geq 0$. Moreover, for all time $t > 0$, $f(t)$ is bounded and belongs to $C^\infty(\mathbb{R}^N_v)$.*

As far as we know, this Proposition was the first regularization effect rigourously observed in the context of the Landau equation. Since the completion of this work, extensions to other potentials have been given in [8]. This supports the conjecture that grazing collisions have a tendency to regularize the solution of the Boltzmann equation (see [23] and the references inside for background on this subject). This is very different to what happens in the context of the Boltzmann equation with cut-off, whose solution usually keeps at most the regularity of the initial data.

We shall now make a simple remark. With obvious notations, we set $Q(f,h) = \mathcal{L}_f \cdot h$. As we have seen in section 2, $\mathcal{L}_f$ depends only on the "gross conditions" of f_0 (its moments up to order two), and on t. This entails

Proposition 5. *(i) The solutions of equation (1) for a given set of gross conditions form a convex set.*
(ii) Let f be a solution of equation (1), and let g satisfy equation (9), such that g has vanishing mass, momentum and pressure tensor, and $f + g \geq 0$. Then $f + g$ is a solution of equation (1).

By definition, the pressure tensor is $\int g v_i v_j$. The proof is immediate. For example, suppose that f and g are two solutions for which those parameters are the same, and take any two nonnegative real numbers λ and μ such that $\lambda + \mu = 1$,

$$\begin{aligned} Q(\lambda f + \mu g, \lambda f + \mu g) &= \lambda Q(\lambda f + \mu g, f) + \mu Q(\lambda f + \mu g, g) \\ &= \lambda Q(f,f) + \mu Q(g,g). \end{aligned}$$

We conclude this section with a very brief discussion of numerical simulations. The only case in which explicit solutions are not available is when $(D_{ij}) \neq 0$, that is, $\int f v_i v_j$ is not always equal to δ_{ij}. But even in the case when $(D_{ij}) = 0$, the intricate form of the solution makes it worth simulating the equation for numerical purposes. The most simple scheme to think of is to simulate first a Fokker-Planck-type diffusion, then a spherical diffusion. This is allowed since $\mathcal{L}_1$ and $\mathcal{L}_2$ commute.

Now, if $(D_{ij}) \neq 0$, we can still apply Trotter's formula, which would yield, if $\mathcal{L}_2$ did not depend on t,

$$e^{t\mathcal{L}} f_0 = \lim_{n\to\infty} \left(e^{\frac{t}{n}\mathcal{L}_1} e^{\frac{t}{n}\mathcal{L}_2} \right)^n f_0,$$

In fact $\mathcal{L}_1$ is a distorted diffusion whose coefficients depend on t. Therefore the preceding formula is not of practical use; it is better to replace in it $\mathcal{L}_1$ by the operator L_1 obtained by freezing the coefficients of $\mathcal{L}_1$ at time 0. Note that L_1 still depends on the matrix (D_{ij}) (which can be assumed to be diagonal) of the function to which it is applied. Therefore all L_1 in Trotter's formula will have different coefficients.

It is possible to implement this scheme in a deterministic way. As both $\mathcal{L}_1$ and $\mathcal{L}_2$ preserve the mass, momentum, energy, positivity, and make the entropy decrease, we can think that the splitting will do the same. We note that $\partial_{ij} f$ should be written $\partial_i(f\partial_j \ln f)$ to obtain a discrete scheme forcing the entropy to decrease [5].

5. Final decomposition of the Landau collision operator

In this very short section, we only sum up the decomposition that was done in the last section. Note that we use the same variable t as in equation (1), while in all the remaining of the paper, we rescale time to divide the collision kernel by $N-1$ so that it coincide with the linear Fokker-Planck operator in the isotropic case.

New form of the collision operator : In a well-chosen orthonormal basis, the Landau equation (1) may be written as

$$\partial_t f = N\nabla\cdot(\nabla f + fv) - \left\{ \sum_i T_i \partial_{ii} f + \nabla\cdot(fv) \right\} + \Delta_\omega f$$

$$= \sum_i (N - T_i)\partial_{ii} f + (N-1)\nabla\cdot(fv) + \Delta_\omega f,$$

where Δ_ω is the Laplace-Beltrami operator, and

$$T_i = \int f v_i^2 = 1 + (T_i(0) - 1)\, e^{-4Nt},$$

$$\int f = 1; \quad \int f v_i = 0; \quad \int f|v|^2 = \sum_i T_i = N; \quad i \neq j \Rightarrow \int f v_i v_j = 0.$$

6. Weak formulations and applications : time-evolution of the moments, Maxwellian tails, energy estimates

Thanks to the results of the preceding section, all the formal computations that we shall do in this section are easily justified, provided that the integrals converge.

Starting from equation (9), we multiply its solution f by a given smooth function $\varphi(v)$ and integrate over $\mathbb{R}^N$. We still assume conditions (8). Then

$$\frac{d}{dt}\int f\varphi = \int \sum_{ij} \overline{a}_{ij}\partial_{ij}f\varphi + N\int f\varphi = \int f <\varphi>, \tag{14}$$

$$<\varphi> = \sum_{ij} \partial_{ij}(\overline{a}_{ij}\varphi) + N\varphi.$$

An easy computation (recall that $\mathcal{L}_2$ is self-adjoint) yields

$$<\varphi> = \mathcal{L}^*\varphi = \sum_{ij} \overline{a}_{ij}\partial_{ij}\varphi - 2\sum_i v_i\partial_i\varphi, \tag{15}$$

$$\overline{a}_{ij} = \left(1 + \frac{|v|^2}{N-1}\right)\delta_{ij} - \frac{v_iv_j}{N-1} - e^{-\alpha t}D_{ij}.$$

It will be useful to give a simplified expression of $<\varphi>$ when φ is radial. Assume $\varphi(v) = \psi(x)$, $x = |v|^2/2$, then, remembering that $\sum_i D_{ii} = 0$, we obtain

$$<\varphi> = \psi'(x)(N-2x) + \psi''(x)\left(2x - e^{-\alpha t}\sum_{ij} D_{ij}\, v_iv_j\right). \tag{16}$$

From now on, we assume that $D_{ij} = D_{ii}\delta_{ij}$. Of course, we recover mass and energy conservation for $\psi = 1, x$: $<1> = 0$ and $<x> = N - 2x$, whence

$$\frac{dE}{dt} = \int f(N-2x) = N - 2\,\frac{N}{2} = 0.$$

Now we can do for the Landau equation the same study as was done in a classical paper by Ikenberry and Truesdell [11] : as we shall see, here we have no need to introduce spherical harmonics. We set

$$M_{2r|\beta} = \int f v^{2r} v^{\beta},$$

with the usual notation $v^\beta = v_1^{\beta_1}\cdots v_N^{\beta_N}$, $v^{2r} = |v|^{2r}$, and the convention that the M with negative index are 0. The motivation for introducing these functions stems from the fact that one cannot obtain directly a closed differential equation for the $M_{0|\beta}$'s. Denoting by k_i the n-uple whose only nonzero coefficient is k, at position i, we have

$$\partial_i(v^{2r}v^\beta) = 2rv^{2(r-1)}v^{\beta+1_i} + \beta_i v^{2r}v^{\beta-1_i},$$

$$\begin{aligned}\partial_{ij}(v^{2r}v^\beta) = (2r)(2r-2)v^{2(r-2)}v^{\beta+1_i+1_j} \\ + \beta_j(2r)v^{2(r-1)}v^{\beta+1_i-1_j} + \beta_i(2r)v^{2(r-1)}v^{\beta+1_j-1_i} \\ + \beta_i\beta_j v^{2r}v^{\beta-1_i-1_j} + \delta_{ij}\left[(2r)v^{2(r-1)}v^\beta - \beta_i v^{2r}v^{\beta-2_i}\right].\end{aligned}$$

Thus we get, after another tedious computation, setting $|\beta| = \sum_i \beta_i$,

$$\begin{aligned}< v^{2r}v^\beta >= - \left[\frac{1}{N-1}\left(|\beta|^2 + (2N-3)|\beta|\right) + 2r\right] v^{2r}v^\beta \\ + 2r(2r-2+2|\beta|+N)v^{2(r-1)}v^\beta \\ + \sum_i(\beta_i^2-\beta_i)v^{2r}v^{\beta-2_i} + \sum_i \frac{(\beta_i^2-\beta_i)}{N-1}v^{2(r+1)}v^{\beta-2_i} \\ -\sum_i e^{-\alpha t}D_{ii}\left((2r)(2r-2)v^{2(r-2)}v^{\beta+2_i} + 2\beta_i(2r)v^{2(r-1)}v^\beta + (\beta_i^2-\beta_i)v^{2r}v^{\beta-2_i}\right).\end{aligned}$$

Using (14), we can express $\frac{d}{dt}M_{2r|\beta}$ as a linear combination of $M_{2r|\beta}$, $M_{2(r+1)|\beta-2_i}$, $M_{2(r-1)|\beta}$, $M_{2r|\beta-2_i}$, $M_{2(r-2)|\beta+2_i}$. We write it symbolically

(17)
$$\frac{d}{dt}M_{2r|\beta} = F(M_{2r|\beta}, M_{2(r+1)|\beta-2_i}, M_{2(r-1)|\beta}, M_{2r|\beta-2_i}, M_{2(r-2)|\beta+2_i}).$$

We remark that all the moments $M_{2r'|\beta'}$ intervening in F are such that $2r' + |\beta'| \le 2r + |\beta|$, with equality only if $r = r'$, $\beta = \beta'$, or $r' = r+1$, $|\beta'| = |\beta|-2$. Thus we obtain a rule analogous to Truesdell's.

Rule : *To establish explicitly the differential equation satisfied by an arbitrary moment of the form $M_{2r|\beta}$, it suffices to write* **all** *the differential equations (17)*
(i) by increasing order of $\gamma = 2r + |\beta|$;
(ii) for a fixed value of γ, by decreasing order of r,
until the desired values of r and β are attained. All these equations are then explicitly solvable one after another.

For $|\beta| + 2r \geq 3$, the coefficient appearing before $M_{2r|\beta}$ in equation (17) is a strictly negative, decreasing function of $|\beta|$ and of r. As soon as $|\beta| \geq 3$, or $|\beta| = 1, 2$ and $r \geq 1$, or $|\beta| = 0$ and $r \geq 2$, it is less than -4. This makes it easy to prove, by a straightforward induction, that each moment returns exponentially fast to its equilibrium value faster than Ce^{-4t}. We note that there is no contradiction in that the coefficient is not always equal to α for $|\beta| = 2$: indeed, we have chosen a basis in which (D_{ij}) is diagonal. We sum up with the

Theorem 6. *(i) Every moment $M_{2r|\beta}(t)$ returns exponentially fast to its equilibrium value, provided that it is finite for $t = 0$.*
(ii) Moreover, $|M_{2r|\beta}(t) - M_{2r|\beta}(\infty)| \leq Ce^{-4t}$ as $t \to \infty$, and in general $|M_{2r|\beta}(t) - M_{2r|\beta}(\infty)| \sim Ce^{-4t}$, for $2r + |\beta| \geq 3$.
(iii) There is no appearance of moments for the Landau equation, that is, no moment ever becomes finite if it is infinite at $t = 0$.

Parts 1 and 3 of the above theorem are true for the Boltzmann equation with Maxwellian potentials : the first part was proven by Truesdell, [11] and the third one is obvious on the basis of Wild's formula. We note in passing that one of Truesdell's conjecture concerning the equations satisfied by the moments relies upon a non-resonance condition ([11], eq. (52.14)) that was proven by Bobylev for quite different purposes ([3], eq. $14.9 - 14.11$).

We emphasize, however, that the second part of the theorem is *false* for the Boltzmann equation. Indeed, Truesdell proved that for the latter, *no moment relaxes more slowly than the moments of order 2.* On the contrary, in the case of the Landau equation, it is easily checked that, for example, if $(D_{ij}) = 0$,

$$\frac{d}{dt} \int f(|v|^2 v_1) = -4 \int f(|v|^2 v_1),$$

so that relaxation occurs at a rate 4 for this moment (to compare with $4N/(N-1)$).

We shall now use formula (14) to investigate the behaviour of

$$\chi_f(K) = \int_{|v|^2/2 \geq K} f \frac{|v|^2}{2}.$$

χ_f "measures" the way the energy is distributed among the velocities, and it is very useful in the study of some properties of kinetic equations, as can be seen in [6] for example. As the function $x 1_{x \geq K}$ is singular, we shall replace it by a smooth approximation.

We consider η a smooth function defined on $[0,\infty)$, such that $0 \leq \eta \leq 1$, $\eta(x) = 0$ if $x \leq 1/2$, $\eta(x) = 1$ if $x \geq 1$, and set

$$\zeta_K(v) = x\eta\left(\frac{x}{K}\right), \quad \theta_f(K) = \int f\zeta_K$$

so that

$$\chi_f(K) \leq \theta_f(K) \leq \chi_f\left(\frac{K}{2}\right).$$

Using (16), we find

$$\frac{d}{dt}\int f\zeta_K = -2\int x\zeta\left(\frac{x}{K}\right)f + N\int \zeta\left(\frac{x}{K}\right)f$$
$$+\int\left((N+4)\frac{x}{K} - 2\frac{x^2}{K} - 2e^{-\alpha t}\sum_{ij} D_{ij}\frac{v_iv_j}{K}\right)\zeta'\left(\frac{x}{K}\right)f$$
$$+\int\left(2\frac{x^2}{K^2} - e^{-\alpha t}\sum_{ij} D_{ij}\frac{v_iv_jx}{K^2}\right)\zeta''\left(\frac{x}{K}\right)f.$$

Now $(x/K)\zeta'(x/K) \leq |\zeta'(x/K)| \leq C\zeta(2x/K)$ (here and elsewhere C will denote any constant depending not on f, but only on the normalization (8)), and

$$\int \zeta\left(2x/K\right)f \leq \frac{C}{K}\int xf,$$

and so on. Estimating separately all terms of this expression, we obtain finally

$$\frac{d}{dt}\theta_f(K) \leq -2\theta_f(K) + \frac{C}{K},$$
$$\theta_f(K,t) \leq \theta_f(K,0)e^{-2t} + \frac{C}{K}e^{-2t}\int_0^t e^{2s}ds \leq \frac{N}{2}e^{-2t} + \frac{C}{K}.$$

Corollary 6.1. *$\chi_f(K,t) \longrightarrow 0$ as $K \to \infty$, $t \to \infty$, uniformly with respect to all f whose mass and energy is fixed (or bounded by a given constant).*

We note that the properties of χ_f are not so good in the Boltzmann context. As a simple illustration of the interest of this corollary, we mention that the following property holds [20].

Theorem 7. *Let (f_n) be a sequence of solutions of Boltzmann or Landau equations for Maxwellian molecules, such that (f_n) exists for the time t_n, $(t_n) \to \infty$, and $\chi_{f_n}(K,t_n)$ goes to 0 as K tends to infinity, uniformly with respect to n. Then, for any fixed time t, $f_n(t)$ tends towards a Maxwellian distribution (in weak-measure sense for example).*

The following consequence indicates that it is actually impossible to solve the backward Landau equation.

Corollary 7.1. ("Liouville theorem for the Landau equation") *Let f be a solution of (1) for $t \in \mathbb{R}$. Then f is a Maxwellian.*

Finally, we shall be interested in the distribution tails for the Landau equation. We recall a definition from [2].

Definition 1. *The temperature tail $T(f)$ of a nonnegative function f is given by*

$$T(f)^{-1} = \sup \left\{ \beta / \int f(v) e^{\beta |v|^2/2} \, dv < \infty \right\}.$$

A distribution function f such that $T(f) < \infty$ is called rapidly decreasing.

Bobylev showed that the space of rapidly decreasing functions was particularly well-suited for the study of the Boltzmann equation with Maxwellian molecules, and constructed a whole theory in that frame. He remarked that, if f is a solution of the Boltzmann equation, then T is an increasing function of t, as can be seen most easily from the inequality $f(t,v) \geq f_0(v) e^{-t}$. This can be interpreted by saying that the distribution tail can only become worse with time (and of course T can be a strictly increasing function). This prevents the convergence towards the equilibrium to hold in L^1 with weight $e^{|v|^2/2}$ if the initial temperature tail is higher than the equilibrium temperature, that is, if $T(f_0) > 1$ in our case.

In this respect too, the Landau equation appears to depart from the Boltzmann equation, as we shall see.

We first treat the case when $(D_{ij}) = 0$. We choose $\psi(x) = e^{\beta x}$ in (16), $\beta > 0$, and compute

$$\frac{d}{dt} \int f\varphi = \beta N \int f\varphi - 2\beta \int x\varphi f + 2\beta^2 \int x\varphi f$$

$$\leq \beta N \int f\varphi + 2\beta(\beta - 1) \int x\varphi f = \beta N \int f\varphi \quad \text{if } \beta < 1. \tag{18}$$

So, if $\int f\varphi$ is finite at time 0, it will always remain finite. Thus, if the initial temperature tail is chosen too high, it does not increase with time, contrary to what happens for the Boltzmann equation.

In fact, we can have an exact estimate of T. Since $e^{\beta |v|^2/2}$ is radial and $\mathcal{L}_2$ is self-adjoint and does not change radial functions, it suffices to consider the solution of the Fokker-Planck equation : $f = M_{\delta(t)} *$

$e^{Nt}f_0(e^t\cdot)$ (remember that $\mathcal{L}_1$ and $\mathcal{L}_2$ commute if $(D_{ij})=0$). Clearly,

$$T(M_\delta(t))=\delta(t)=1-e^{-2t};\quad T(f_0(e^t\cdot))=T(f_0)e^{-2t}.$$

Lemma 1. *Let $\gamma\geq 0$, and let g be a nonnegative L^1 function. Then*

$$T(M_\gamma * g)=T(g)+\gamma.$$

Proof. The proof just consists in making rigorous the heuristic idea that $M_\gamma * M_{-\epsilon}=M_{\gamma-\epsilon}$ for $\epsilon>\gamma$. We choose $\Phi(v)$ a radial cut-off function of the form $1_{|v|<K}$. Then, for any $\epsilon>T(g)+\gamma$,

$$\int (M_\gamma * g)e^{\frac{|v|^2}{2\epsilon}}\Phi(v)\,dv=\int e^{-\frac{v_*^2}{2\gamma}}e^{\frac{|v-v_*|^2}{2\epsilon}}\Phi(v-v_*)g(v)\,dv\,dv_*.$$

Since

$$-\frac{|v_*|^2}{2\epsilon}+\frac{|v-v_*|^2}{2\epsilon}=$$

$$-\left(v_*\sqrt{\frac{1}{2\gamma}-\frac{1}{2\epsilon}}+\frac{v}{2\epsilon}\frac{1}{\sqrt{\frac{1}{2\gamma}-\frac{1}{2\epsilon}}}\right)^2+\frac{|v|^2}{4\epsilon^2}\left(\frac{1}{2\gamma}-\frac{1}{2\epsilon}\right)^{-1}+\frac{|v|^2}{2\epsilon},$$

this expression is equal to

$$\int e^{-\left(v_*\sqrt{\frac{1}{2\gamma}-\frac{1}{2\epsilon}}+\frac{v}{2\epsilon}\frac{1}{\sqrt{\frac{1}{2\gamma}-\frac{1}{2\epsilon}}}\right)^2}e^{\frac{|v|^2}{2(\epsilon-\gamma)}}\Phi(v-v_*)g(v)\,dv\,dv_*$$

$$\leq C_{\epsilon,\gamma}\int e^{\frac{|v|^2}{2(\epsilon-\gamma)}}g(v)\,dv\,dv_*.$$

We now apply the monotone convergence theorem, letting K go to infinity : this yields $\int e^{\frac{|v|^2}{\epsilon}}(M_\gamma * g)<\infty$, and thus $T(M_\gamma * g)\leq T(g)+\gamma$. The reverse inequality is obtained in the same manner. □

Corollary 7.2. $T(f_t)=1+(T(f_0)-1)\,e^{-2t}$.

Thus T converges monotonically to 1; this should be compared to the Boltzmann equation, in the frame of which Bobylev proved : $1-e^{-\lambda t}\leq T(f)\leq a$, where a is some constant depending on f_0, that can be as large as we wish.

Unfortunately, this monotone behaviour is not always true when $(D_{ij})\neq 0$: as the diffusion is faster along some of the directions, the temperature tail can increase. Using the change of variables mentioned in section 4, we can control it. Or we can go back to (18) : there will be an additional term

$$-e^{-\alpha t}\beta^2\sum_i D_{ii}\int v_i^2 f\leq 2\beta^2 e^{-\alpha t}\int x\varphi f,$$

(we have used $|D_{ii}| \leq 1$) and we see that the conclusion still holds if $(\beta^2(2+2e^{-\alpha t}) - 2\beta) \leq 0$, so that at every time when

$$T(f_t) \geq 1 + e^{-\alpha t},$$

T has to decrease. This proves that T converges to 1.

7. Positivity

Suppose $f_0 \geq CM_\theta$, for some $C > 0$, $\theta > 0$. We shall apply the maximum principle to prove that f is always bounded from below by a Maxwellian function.

We set

$$\varphi(t) = e^{-\delta(t)\frac{|v|^2}{2}} e^{-\beta(t)},$$

where δ and β will be chosen later. Since f is a solution of $\partial_t f - \mathcal{L} f = 0$, it is sufficient for f to be bounded from below by φ, that $\partial_t \varphi - \mathcal{L}\varphi \leq 0$, ie

$$-\partial_t\varphi + \Delta\varphi + v\cdot\nabla\varphi + N\varphi - e^{-\alpha t}\sum_i D_{ii}\partial_{ii}\varphi$$

$$= \left(\beta' - N(\delta - 1) + \left(\frac{\delta'}{2} + \delta^2 - \delta\right)|v|^2 - e^{-\alpha t}\delta^2 \sum_i D_{ii} v_i^2\right)\varphi \geq 0.$$

Since

$$\left|\sum_i D_{ii} v_i^2\right| \leq (\sup|D_{ii}|)|v|^2 \leq |v|^2,$$

it is sufficient that for all time t,

$$\begin{cases} \beta'(t) \geq N(\delta(t) - 1) \\[2ex] \dfrac{\delta'(t)}{2} \geq \delta(t) - \delta^2(t) + e^{-\alpha t}\delta^2(t) \end{cases}$$

We assume $\delta > 1$ and set $\delta = 1 + \gamma$. For the second inequality to be satisfied, it is sufficient that

$$\frac{\gamma'}{2} \geq -\gamma + e^{-\alpha t}(1 + 2\gamma),$$

so we choose γ to be the solution of the differential equation

$$\gamma' = -2\gamma + 2e^{-\alpha t}(1 + 2\gamma),$$

with initial condition $\gamma(0) = \theta - 1$ (we can always assume that $\theta > 1$). We note that γ is always nonnegative (if not, there would be a time t such that $\gamma(t) = 0$ and $\gamma'(t) < 0$, but then $\gamma'(t) = 2e^{-\alpha t}$). For t large enough, $\gamma' \leq -\gamma + 2e^{-\alpha t}$. Since $\alpha > 2$, this entails that γ converges exponentially fast to 0 as $t \to \infty$, and we can then choose $\beta' \geq N\gamma$ such that $\int_0^\infty |\beta'(t)|\, dt < \infty$. This proves in the end the

Theorem 8. *If $f_0 \geq CM_\theta$ $(C > 0,\ \theta > 0)$, then there exists $C' > 0$ such that*

$$f(t,v) \geq C' e^{-\delta(t)\frac{|v|^2}{2}},$$

$$\delta(t) \longrightarrow 1 \quad \textit{as } t \to \infty.$$

Now, if no lower bound is available on f_0, then the case where $(D_{ij}) = 0$ is still immediate : it suffices to consider the Fokker-Planck equation. It is proven in [7] for instance that for this equation, Maxwellian lower bounds immediately appear; then one can apply Theorem 8.

The general case is slightly less easy. We shall limit ourselves to the following proposition, that has counterparts for pretty general potentials, and is therefore proved elsewhere [8] :

Proposition 9. *For all $t > 0$, there exist C_t and δ_t such that*

$$f(t) \geq C_t e^{-\delta_t \frac{|v|^2}{2}}.$$

This proposition entails in particular that the conclusion of Theorem 8 is valid in all cases as soon as $t > 0$.

8. Long-time behaviour

We set $H(f) = \int f \ln f$. The estimates proven above :

$$C_1 e^{-\delta |v|^2/2} \leq f \leq C_2$$

for any time $t > 0$ prove that *for any time $t > 0$, H is well-defined.* It is then classical that

$$\frac{d}{dt} H(f(t)) = -\int a_{ij}(v - v_*) f f_* \left(\partial_i \ln f(v) - \partial_i \ln f(v_*)\right) \\ \left(\partial_j \ln f(v) - \partial_j \ln f(v_*)\right) \leq 0.$$

This can also be checked by multipying the equation $\partial f = \mathcal{L} f$ by $\varphi = \ln f$. We see that both contributions from $\mathcal{L}_1$ and $\mathcal{L}_2$ are nonpositive. Indeed (with obvious notations),

$$\int r^{N-1} \Delta_\omega f(r,\omega) \ln f(r,\omega)\, dr\, d\omega = -\int r^{N-1} \frac{\nabla_\omega f \nabla_\omega f}{f} dr\, d\omega \leq 0;$$

or alternatively,

$$\int \Big[(|v|^2\delta_{ij} - v_iv_j)\partial_{ij}f - (N-1)v_i\partial_i f\Big] \ln f \, dv$$
$$= -\int (|v|^2\delta_{ij} - v_iv_j)\frac{\partial_i f \partial_j f}{f} \le 0.$$

as for $\mathcal{L}_1$, we can for example change variables as in section 4 : $v = S\tilde{v}$,

$$\int \mathcal{L}_1 f(v) \ln f(v) \, dv = |S| \int L\tilde{f}(\tilde{v}) \ln \tilde{f}(\tilde{v}) \, d\tilde{v} \le 0$$

since $|S| = \Pi(1 + D_{ii}e^{-\alpha t}) \ge 0$.

Remark. From the inequality

$$\sum_i \lambda_i = 0 \Rightarrow 1 \ge \Pi(1 + \lambda_i e^{-\alpha t}) \ge \Pi(1 + \lambda_i) > 0$$

(easy to obtain by Taylor formulae) we see that the decrease of entropy is less rapid that in the case $(D_{ij}) = 0$.

The entropy decrease can be 0 only if f is radial, so that $(D_{ij}) = 0$, and if it is 0 for the linear Fokker-Planck operator, that is, f is Maxwellian.

It is straightforward to adapt to this case all the techniques known for Boltzmann equation, based on the entropy production, to prove that f goes to equilibrium as $t \to \infty$. This occurs *even if the entropy does not exist at time 0.* We mention that the return to equilibrium also holds for Boltzmann equation with Maxwellian molecules even if the entropy does not exist at time 0; it is likely that the same is true for hard potentials [24] (see [1] for complete proofs in the case of hard spheres).

The entropy is not the only Lyapunov functional associated with the Boltzmann equation for Maxwellian potential. Some years ago, Bobylev and Toscani found a sufficient condition for an isotropic convex functional I to be decreasing with time along the solutions of Boltzmann equation, [4] in the two-dimensional or three-dimensional axisymmetric case :

$$I(f_\lambda * g_\mu) \le \lambda I(f) + \mu I(g), \tag{19}$$

where $f_\lambda = \lambda^{-N/2} f(\cdot/\sqrt{\lambda})$, $\lambda, \mu > 0$, $\lambda + \mu = 1$.

Let I such a functional, we fix some initial data f_0 and $t > 0$, and consider f the solution of equation (1) on $[0, t]$. We consider any Boltzmann equation with Maxwellian kernel, it is then possible to do the asymptotics of grazing collisions for this kernel [21], so that we can produce a sequence of solutions g^ε of the Boltzmann equation (with initial data f_0) converging (up to extraction) on $[0,t]$ to a solution of the Landau equation. The uniqueness (of the weak solution) enables us to identify f with this limit. Now, due to the convexity of I,

$$I(f_t) \leq \underline{\lim}\, I(g_t^\varepsilon) \leq I(f_0).$$

Proposition 10. *The condition (19) is a sufficient condition for I to be decreasing along the solutions of Landau equation, for $N = 2$ or $N = 3$ in the axisymmetric case.*

This is true in particular for the Fisher information (or Linnik functional), and the Tanaka functional. It entails better results of convergence to equilibrium, as in the works by Toscani. The particular case when $(D_{ij}) = 0$ can be investigated with much more details, since explicit solutions are then available. See [17] for a related study.

Remark. We shall prove elsewhere [22] that the Fisher information is in fact always decaying with time, whatever the dimension.

We conclude with a precise study of the rate of the decay to equilibrium. It turns out that this latter is much better controlled in the case of the Landau equation than in the Boltzmann case. Indeed, we recall Bobylev's results [3] : for the Boltzmann equation with Maxwellian potential, if the initial data is rapidly decreasing, then the solution goes to a Maxwellian as $e^{-\mu t}$, where μ is the first eigenvalue of the linearized collision operator. On the other hand, if f_0 is allowed to decrease slowly at infinity, the decay to equilibrium can be slower than $e^{-\gamma t}$, for *any* $\gamma > 0$.

This phenomenon was suggested to Bobylev by the existence of nonzero eigenvalues as small as we wish, in slowly-decaying context. In fact (with the notations of section 3), the eigenvalues are of the form

$$\lambda_\ell(p) = \frac{\pi}{2}\int_0^\pi d\theta \zeta(\theta) \left\{ \cos^p \frac{\theta}{2} P_\ell \left(\cos\frac{\theta}{2} \right) + \sin^p \frac{\theta}{2} P_\ell \left(\sin\frac{\theta}{2} \right) - 1 - \delta_{\ell 0} - \delta_{p0} \right\},$$

with ℓ integer, P_ℓ denotes the ℓ^{th} Legendre polynomial, and $Re(p) \geq 2$. For $p = 2 + \varepsilon$, $\varepsilon << 1$, $\lambda_\ell(p)$ is very small ([3], p.137).

However, in the asymptotics of the grazing collisions, we let formally B tend to δ'', where δ is the Dirac distribution concentrated at 0, and these eigenvalues tend (up to a multiplicative factor) towards $-(\pi/8)(p + P'_\ell(1)) = -(\pi/8)[p + \ell(\ell+1)/2]$, so that they are no longer close to 0. This suggests that the slowly-decaying solutions disappear for the Landau equation. In fact we can show that the rate of decay to equilibrium is indeed bounded away from 0.

The simplest method to prove this, in our opinion, is to adapt the argument given by Toscani [17] for the Fokker-Planck equation. For all i, $D_{ii} \leq 1$; let us denote by Δ_ω the Laplace-Beltrami operator, then

$$\partial_t f = \big(1 - e^{-\alpha t}\big) L f + \frac{1}{N-1}\Delta_\omega f + e^{-\alpha t}\nabla\cdot(fv) + e^{-\alpha t}\sum_i (1 - D_{ii})\partial_{ii} f;$$

since

$$\int (1 - D_{ii})\partial_{ii} f \, \log f = -\int (1 - D_{ii}) \frac{\partial_i f \partial_i f}{f} \leq 0,$$

and by an integration by parts,

$$\int \nabla\cdot(fv)\log f = N;$$

so that we can estimate the entropy dissipation by

$$\frac{d}{dt} H(t) \leq -\big(1 - e^{-\alpha t}\big)\left(\int \frac{\nabla f \nabla f}{f} - N\right) + N e^{-\alpha t}.$$

Now, we use the logarithmic Sobolev inequality, in the form

$$\int f \log f - \int \mathcal{M}\log\mathcal{M} \leq \frac{1}{2}\left(\int \frac{\nabla f\nabla f}{f} - N\right)$$

to get a differential inequation for $D(t)$, the relative entropy of $f(t)$ with respect to the associated Maxwellian,

$$\frac{d}{dt} D(t) \leq -2(1 - e^{-\alpha t}) D(t) + N e^{-\alpha t}. \tag{20}$$

Since $\alpha > 2$, this entails that f converges exponentially in relative entropy at a rate better than $2(1-\varepsilon)$, for all $\varepsilon > 0$, as soon as t is large enough. Using the Csiszar-Kullback inequality

$$\|f - \mathcal{M}\|_{L^1} \leq \sqrt{2D(t)},$$

we finally obtain the following

Theorem 11. *As t goes to infinity,*

$$\underline{\lim}\left(-\frac{1}{t}\right)\log\|f(t) - \mathcal{M}\|_{L^1} \geq 1.$$

Moreover, thanks to (20), the relative entropy of f with respect to the equilibrium distribution can be readily estimated using only the relative entropy of the initial data.

Remark. Here we have used in a crucial way the fact that the behaviour of the second-order moments is a priori known. This relies strongly on the assumption that the potential is of Maxwellian type. Compare with [9].

9. Self-similar solutions

We shall exhibit a solution whose decay is exactly like $e^{-\gamma t}$, for any real number $\gamma \geq 2$. We look for a "self-similar" solution of equation (9), that is, of the form $\theta(t)g(v)$. It is sufficient that :

$$\theta'(t) = -\gamma\theta(t), \quad \mathcal{L}g = -\gamma g.$$

Assuming g to be radial, we just have to look for an eigenfunction of the Fokker-Planck operator. Turning to Fourier variables, and setting $\hat{g} = \hat{\mathcal{M}}\hat{\varphi}$, we obtain

$$\xi\partial_\xi\hat{\varphi} = \gamma\hat{\varphi},$$

so that, since $\hat{\varphi}$ is radial, $\hat{\varphi} = \theta|\xi|^\gamma$ for some positive constant θ, whence, by inverting the Fourier transform we obtain, up to a constant depending only on γ and N (Cf [3], p.223),

$$g(v) = \mathcal{M}\theta\,{}_1F_1\left(-\frac{\gamma}{2}, \frac{N}{2}, \frac{|v|^2}{2}\right),$$

where ${}_1F_1$ denotes the degenerate hypergeometric function (see Appendix A for more details).

We note that g has vanishing mass, momentum and energy (the first derivatives of its Fourier transform vanish), and we have estimates on its asymptotic behaviour thanks to those on ${}_1F_1$ (see for instance [13]) : with $x = |v|^2/2$,

$$g(v) \sim Ce^{-x}x^{-\frac{N+\gamma}{2}} = \frac{C\mathcal{M}(v)}{|v|^{N+\gamma}} \qquad \text{as } |v| \to \infty,$$

for $\gamma/2 \notin \mathbb{N}$; this shows that we must require $\gamma > 2$ for the energy to be finite, and also that $e^{-x} + \theta y(x)$ is nonnegative for sufficiently small θ.

Therefore we can apply the second part of Proposition 5 and conclude that

$$f(t,v) = \mathcal{M}(v) + \theta e^{-\gamma t}g(v)$$

is a solution of the Landau equation. Bobylev had already noted that such initial conditions entailed a slow decreasing of f to its equilibrium in the frame of the Boltzmann equation. Here, the decay is not so slow, in accordance with the results of section 8.

Finally, we note that Lemou's particular solutions (and Bobylev's simplest solution as well) can be recovered in that way : indeed, when $\gamma/2 \in \mathbb{N}$, the degenerate hypergeometric function reduces to a Laguerre polynomial for the weight function $e^{-x}\sqrt{x}$. For example, the solution cited in section 3 is obtained with $\gamma = 4$, and corresponds to the second

Laguerre polynomial. One can also mix different values of γ thanks to Proposition 5.

Appendix A. Inverse Fourier Transform of $e^{-|\xi|^2/2}|\xi|^\lambda$

This inverse Fourier transform (used by Bobylev [3] for the three-dimensional case) is not always found in tables, so we briefly indicate how to compute it.

We set $\varphi(r) = e^{-r^2/2}r^\lambda$. We first reduce the problem to a one-dimensional one with a polar change of coordinates, $\xi = r\omega$,

$$\int e^{iv\cdot\xi}\varphi(|\xi|)d\xi = \int e^{iv\cdot\xi}\varphi(r)r^{N-1}dr\,d\omega.$$

It is classical that, by symmetry,

$$\int e^{iv\cdot\xi}d\omega = \int e^{i|v|r\omega_1}d\omega = |S^{N-2}|\int_{-1}^{1} d\omega_1(1-\omega_1^2)^{\frac{N-3}{2}}\cos(|v|r\omega_1),$$

where $|S^{N-2}| = 2\pi^{\frac{N-1}{2}}/\Gamma(\frac{N-1}{2})$ is the volume of the $(N-2)$-dimensional unit sphere. Now the integral is expressed thanks to a Bessel function of the first kind,

$$\int_{-1}^{1} d\omega_1(1-\omega_1^2)^{\frac{N-3}{2}}\cos(|v|r\omega_1) = \sqrt{\pi}\,\Gamma\left(\frac{N-1}{2}\right)2^{\frac{N-2}{2}}\frac{1}{(|v|r)^{\frac{N-2}{2}}}J_{\frac{N-2}{2}}(|v|r).$$

Thus the N-dimensional Fourier transform reduces to a so-called one-dimensional Hankel transform. Taking as new variable $x = |v|^2/2$, we find that, up to a multiplicative constant depending only on N, the unknown function is

$$y(x) = \frac{1}{x^{\frac{N-2}{4}}}\int_0^\infty J_{\frac{N-2}{2}}(\sqrt{2x}\,r)\varphi(r)r^{\frac{N}{2}}dr.$$

This integral can be found in [10] (formula 6.631 – unfortunately, we came across this reference only after doing all the computations !). In brief, to compute it, we look for a differential equation checked by y. To that purpose, we differentiate y two times with respect to x. Using the Bessel differential equation in the form

$$2xr^2J''(\sqrt{2x}\,r) + \sqrt{2x}rJ'(\sqrt{2x}\,r) + \left[2xr^2 - \frac{(N-2)^2}{4}\right]J(\sqrt{2x}\,r) = 0,$$

and an integration by parts

$$\int_0^\infty J'(\sqrt{2x}\,r)\varphi(r)r^{\frac{N+2}{2}}dr = \frac{1}{\sqrt{2x}}\int_0^\infty J(\sqrt{2x}\,r)\varphi(r)r^{\frac{N+4}{2}}dr$$

$$-\left(\frac{N+2}{2}+\lambda\right)\frac{1}{\sqrt{2x}}\int_0^\infty J(\sqrt{2x}\,r)\varphi(r)r^{\frac{N}{2}},$$

it is possible to express y' and y'' as a linear combination of y and $\int_0^\infty J(\sqrt{2x}r)\varphi(r)r^{\frac{N+4}{2}}\,dr$. Eliminating this last function yields

$$xy'' + y'\left(\frac{N}{2}+x\right) + y\left(\frac{N+\lambda}{2}\right) = 0.$$

This, combined with elementary estimates of y, entails that (up to a multiplicative constant)

$$y(x) = e^{-x}\,{}_1F_1\left(-\frac{\lambda}{2},\frac{N}{2},x\right).$$

We recall that ${}_1F_1(\alpha,\gamma,z)$ is one particular solution of the differential equation

$$zF'' + (\gamma - z)F' - \alpha F = 0.$$

For other equivalent definitions and properties of the degenerate hypergeometric function, we refer to [13].

Appendix B. The Nikolskii Transform

The Nikolskii transform [12] enables to obtain a particular solution of the inhomogeneous Boltzmann equation, starting from a solution of the homogeneous Boltzmann equation. We shall give here the straightforward adaptation for the Landau equation (with Maxwellian potential or not).

We rewrite the Landau operator in the general case

$$Q(f,f) = \int dv_*\,a(v-v_*)\Bigg(f(v_*)\nabla f(v) - f(v)\nabla_* f(v_*)\Bigg),$$

where $a(v-v_*)$ has the same homogeneity than $|v-v_*|^{\gamma+2}$. Let us set $f(v) = F(tv-x)$, $t>0$: then $\nabla f(v) = t\nabla F(tv-x)$ and after a change of variables we obtain

$$Q(f,f)(v) = \frac{1}{t^{N+\gamma+1}}Q(F,F)(tv-x).$$

We can seek a solution of the inhomogeneous Landau equation in the self-similar form

$$f(t,x,v) = F\left(\chi(t), tv-x\right) \equiv F(\theta,w)$$

Putting this expression into

$$\partial_t f + v\cdot\nabla_x f = Q(f,f)$$

yields

$$\chi'(t)\frac{\partial F}{\partial \theta}(\theta, w) = \frac{1}{t^{N+\gamma+1}} Q(F,F)(\theta, w).$$

Thus, it suffices to choose F solution of the homogeneous equation, and

$$\chi(t) = D - \left(\frac{1}{N+\gamma}\right)\frac{1}{t^{N+\gamma}}.$$

In particular, for Maxwellian potentials, $\gamma = 0$, and

$$f(t,x,v) = F\left(D - \frac{1}{Nt^N}, tv - x\right).$$

Applying this transformation to the particular solution found in section 8,

$$F(\theta, w) = \mathcal{M}(w)\left(1 + e^{-\lambda\theta} g(w)\right),$$

we get

$$\begin{aligned} f(t,x,v) &= \mathcal{M}(tv-x)[1 + e^{-\lambda\chi(t)} g(tv-x)] \\ &= \mathcal{M}(tv-x) + e^{-\lambda\chi(t)} G(tv-x). \end{aligned}$$

Here we have set $G = \mathcal{M}g$. This is a solution (with infinite mass) of the inhomogeneous Landau equation. It is simply the sum of a travelling Maxwellian and a "travelling perturbation". We note that the asymptotic state is not an equilibrium state, but $(\mathcal{M}+CG)(tv-x)$, that is, a travelling asymptotic state that is not solution of the Landau equation (in fact, f goes to 0 in L^1_{loc} as t goes to infinity).

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